Cluster 03 – Data-driven decision making

Elena's six steps as one cycle · each of us enters at a different step · shared themes on the edges

IN PLAIN WORDS

Someone has to choose between options: a tree, a route, a planting. Data predicts what each option does before it exists, but data does not say which result is better: the model does. **Cities decide with data, but what counts as "correct" is a choice.** (Jaton 2021)

? Could the people affected redefine what counts as correct?

The average vs the single case
- personas from labels → each person
- a city table → each tree position
- the same UTCI engine: Infrared.city

01 Define
what counts as correct? · who is it for?
- anticipated sensory comfort, from a photo
- what the rules allow
- better than today, for the water it costs

evaluate and learn: back to Define

? After building, does anyone check the prediction held?

06 Act
what does a decider see? · better than what?
- a personal map of routes
- a species for each tree to replace
- comfort and water per scenario

02 Collect data
whose data, how much? · at what grain?
- 30–40 people, 20 photos
- every street tree: inventory + LiDAR
- a few points, 10:00–15:00

? Who is missing: night, winter, people who stay home?

GIOVANNI · enters at 06 Act

Which tree replaces which tree, when a project must replace trees?

optimal against the rules · validated case: the plane tree (Platanus)

Methodological gap no method picks species per tree position on heat and runoff together, inside a city's replacement rules · runoff: do not worsen

Same index, same points, same hours
- UTCI, scenarios S0–S4
- UTCI, scenarios E0–E3

Water: kept or spent?
- runoff held back by crowns
- litres per season

A limit declared in advance
- gain shrinks as storms grow
- humidity can cancel the cooling

ELENA · enters at 01 Define

How will a place feel to my senses?

optimal against the individual

Empirical gap no test of whether a persona matches one person's own sensory comfort ratings of public space

DATA-DRIVEN DECISION

WHICH NUMBER DECIDES?

usually: one average person, one target

05 Decide
which number wins? · who decides?
- a personal comfort score
- UTCI; runoff must not worsen
- UTCI next to litres

03 Clean and prepare
which features stand in? · what is lost?
- 16 visible elements per sense
- crown size, leaf area
- 47 native plants audited

? When comfort and water disagree, who weighs them?

04 Analyse
which model, for whom? · checked against?
- a vision-language persona
- shade × leaf area; runoff
- UTCI: vegetation vs water

? What is dropped as noise, and who calls it noise?

"Cooler" ≠ "more comfortable"
- perceived comfort
- thermal comfort, not air temperature

? Can a model of averages stand in for one person or one site?

DAVID · enters at 05 Decide

Does water add walking comfort, for the water it costs?

optimal against the water the site has · Tobia, Colombia

Empirical gap nobody tests the same native planting with and without water, with UTCI and litres

EVALUATE AND LEARN · WHAT EACH GIVES BACK

Elena "Rate this place" feeds the persona; participants check the findings

Giovanni the city's rule-following replacement becomes the baseline; the cost of the 15% cap

David a seasonal water budget next to every cooling claim